

Pratiush Karki

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TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks: React, Next.js, FastAPI, Node.js, Express, Streamlit

AI/ML: LangChain, RAG, ChromaDB, LLaMA, Ollama, scikit-learn

Data: PostgreSQL, Redis, Firebase, pandas, NumPy

Tools/Cloud: Git, GitHub Actions, VS Code, Docker, AWS EC2, Figma

EXPERIENCE

TaxCurb System · Software Engineer

Dec 2025 - Present

- Built real-time risk identification and mitigation into the compliance pipeline by gating deployments through CI/CD-integrated verification and validation, ensuring 100% of code pushes pass compliance-detection accuracy checks before reaching production.
- Designed backend system architecture using Node.js, Express, and Firebase Admin SDK with secure server-side credential isolation, token-based authentication, rate limiting, and structured input validation to ensure reliability and system robustness.
- Built data pipelines to transform sparse, user-generated inputs into normalized, document-ready outputs, strengthening understanding of system representation learning under partially observed data environments.

New York Jets · Data Analyst

Jan 2026 - May 2026

- Led a 5-person student analytics team that identified an underserved college-aged fan segment representing 62% of surveyed respondents by analyzing historical Jets data and survey responses from ~1,000 target consumers.
- Collaborated with the marketing team and came up with three growth strategies pitched to Jets executives, sizing up to ~\$110K in game-day spend per 1,000 students converted.

Instructure · Startup & Innovation Studio · Caldwell, NJ

Spring 2026

- Built a 0-to-1 product, MicroJournal, with a team of 4 in a startup studio run by Instructure's CTO and CPO, leveraging AI tools to accelerate development and testing while following best SWE practices and feedback from the Instructure engineering team.
- Validated demand with a public waitlist and submitted the final product to the App Store (iOS).

Caldwell University · Systems Assistant & Student Tutor · Caldwell, NJ

- Peer Tutor** · Nov 2024 - Feb 2025 · Tutored course material and clarified complex concepts across multiple courses: Data Structures, Algorithms, Front-End, and Full-Stack Web Development, impacting up to 1,500 students.
- System Assistant** · Feb 2023 - Dec 2023 · Supervised audio-visual systems for major campus events with 100+ attendees, and troubleshoot AV failures in real time under live-audience pressure, keeping output uninterrupted.

PROJECTS

Lucid Tone: Real-Time Background Music Generation Platform | Python, FastAPI, WebSockets, Redis, React, WASM

- Developed a Markov-chain music engine and a rendering pipeline capable of generating audio ~100x faster than real time for smooth generative music playback.
- Managed persistent multi-session WebSocket connections with Redis-backed shared state, buffered audio delivery, and Opus encoding to reduce network bandwidth and egress.

Fully Local RAG Agent | Docker, Python, ChromaDB, Ollama, Streamlit, AWS EC2

- Built a fully local AI agent using retrieval-augmented generation (RAG) over an unstructured document corpus, returning grounded answers traceable to their source passages with no data leaving the machine.
- Containerized with Docker and deployed to AWS EC2. Built during a hackathon with a three-person team on a Git workflow with code review.

NILINK: Platform Connecting Brands and Athletes for NIL DEALS | Next.js, TypeScript, PostgreSQL, Tailwind

- Shipped a two-sided marketplace covering the full name-image-likeness deal path: campaign discovery, applications, offers, contracts, and execution. Demoed and judged at the 2026 Capstone Showcase.
- Modeled the deal lifecycle in PostgreSQL with row-level security so each party's commercial terms stay private from the other, then held the same boundary in application code through role-aware routing and protected API routes.

RESEARCH

AI Sycophancy and LLM Reliability · 2026

jstor.org/stable/community.42398302

- Modeled why LLMs tuned on RLHF drift toward sycophancy, and traced the five-stage loop vulnerable users are susceptible to: emotional trigger, biased query, sycophantic response, flawed belief update, amplification.
- Defined boundary conditions and falsifiable predictions, then specified a four-tier mitigation framework across training, product guardrails, and interface design. Applies directly to LLM evaluation, alignment, and trust and safety.

EDUCATION

3.85 GPA · B.S. Computer Science · Business Analytics Minor · Honors Student

Caldwell University

Activities & Awards: Co-founder & Secretary, Computer Science Club, University Choir Singer, Dean's List (8x), Recognition Award